COMPANY CONFIDENTIAL

FSD Headquarters
Rockville, Maryland
December 9, 1960

Memorandum To: Mr. B. G. Oldfield

SUBJECT: FARMBOY Contract

Introduction:

At your suggestion this past week I participated in a brief review and evaluation of the FARMBOY Contract. Particular emphasis was placed on the alpha-language program, since IBM is currently facing a competitive situation in this area.

During the review, Dr. D. Sayre and I met with the following people:

Dr. S. G. Campbell	Mr. Jack Parsons
Dr. P. S. Herwitz	Mr. S. W. Dunwell
Miss Fan Allen	Mr. Carl Gerbesch
Mr. Dick May	Mr. George Grover

Dr. F. H. Brannin

We also had the benefit of marketing opinions from Mr. C.R. Green and Mr. Mike Riley.

General:

The STRETCH Programming Group has the responsibility for developing several classes of programs. One group includes the basic sub-routines for program preparation (e.g., STRAP), debugging, and control of the computer. Another group includes the usual commercial package of FORTRAN and (later) COBOL. The group under consideration here includes programs being paid for by the customer under the FARMBOY Contract. These include: The assembly program (HAP), various debugging routines, and the problem-oriented alpha language.

General (continued):

The STRETCH Programming Group has laid out a procedure by which they hope to minimize the total effort for alpha and FORTRAN by using an intermediate language referred to as "Autocoder", or "Macro Generator" which takes the output of alpha or FORTRAN and prepares symbolic statements for input to the STRAP assembly program.

A controversy has arisen over this autocoder approach with the FARMBOY customer, because he feels that by paying for it he is subsidizing IBM's commercial effort in which he has no interest. (It is interesting to note: a similar concern was expressed this week by one of the AEC customers. He felt that IBM was trying to please the FARMBOY customer to the detriment of the commercial FORTRAN program!) The actual fact seems to be that autocoder is more aimed toward FARMBOY-type (or COBOL type) problems than it is toward scientific FORTRAN problems. The present financial estimate is that FARMBOY will cover about two-thirds of the cost of autocoder.

Observations:

1. The management of the STRETCH Programming Group, and of FARMBOY in particular, appears to be of good quality technically and personnel-wise. The only criticism we can make is that there has been a little laxness in keeping schedules current.

2. The technical approach of using several layers of subroutinization in the production of alpha is complicated but basically sound. This approach will take longer to plan and code, but should take less time to debug and modify later.

3. The team of programmers on FARMBOY appear to be bright and interested in their work. Most of them are relatively inexperienced but, with proper supervision, should be able to do the job. The number of people fully cleared for the project is limited. The ones who are not cleared will continue to work under a handicap.

4. The schedule shown to us is tough but reasonable. Not much "fat" is available to cover excessive machine down time nor other such "act-of-god" delays. The only cushion available is the inevitable one of working everyone more and more overtime hours.

Recommendations:

1. The STRETCH Programming Group should make out and keep up its schedules with more precision. In particular, it would be especially valuable to estimate many specific events or milestones so that anyone can tell how the schedule is coming. A procedure to estimate percentage completed such as that used by Mr. W. C. L. Stone on 438L is recommended. The PERT system, used by the Navy, is more complicated but might give some guidance.

2. IBM should seriously consider not charging any of the autocoder to FARMBOY. This would serve two purposes: (1) It would remove the customer's objection to sponsoring any commercial development, and (2) It would free the STRETCH Programming Group from having to get customer approval for every decision concerning autocoder.

3. There should be another technical review of FARMBOY about March 1, 1961. It is particularly important that Dr. D. Sayre participate in the review.

H. G. Kolsky
Senior Engineer

HGK/del

bcc: S. G. Campbell
N. P. Edwards
D. Sayre
G. Cramer